

# CStats - Lab #4

March 17, 2026

## Exercise 1.

A computer ANOVA output is shown in Table 1. Fill in the blanks. You may give bounds on the  $p$ -value.

| One-way ANOVA |    |        |    |   |   |
|---------------|----|--------|----|---|---|
| Source        | DF | SS     | MS | F | p |
| Factor        | 3  | 36.15  | ?  | ? | ? |
| Error         | ?  | ?      | ?  |   |   |
| Total         | 19 | 196.04 |    |   |   |

Table 1: One-way ANOVA Table

## Exercise 2.

The engineer is interested in determining if the RF power setting affects the etch rate, and she has run a completely randomized experiment with four levels of RF power and five replicates. The data are shown in Table 2. Construct the ANOVA table and test the null hypothesis  $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$  against the alternative  $H_1$ : *some means are different*, for the significance level  $\alpha = 0.05$ .

| RF Power<br>(W) | Observed Etch Rate ( $\text{\AA}/\text{min}$ ) |     |     |     |     | Totals            | Averages                |
|-----------------|------------------------------------------------|-----|-----|-----|-----|-------------------|-------------------------|
|                 | 1                                              | 2   | 3   | 4   | 5   | $y_{i.}$          | $\bar{y}_{i.}$          |
| 160             | 575                                            | 542 | 530 | 539 | 570 | 2756              | 551.2                   |
| 180             | 565                                            | 593 | 590 | 579 | 610 | 2937              | 587.4                   |
| 200             | 600                                            | 651 | 610 | 637 | 629 | 3127              | 625.4                   |
| 220             | 725                                            | 700 | 715 | 685 | 710 | 3535              | 707.0                   |
|                 |                                                |     |     |     |     | $y_{..} = 12,355$ | $\bar{y}_{..} = 617.75$ |

Table 2: The Plasma Etching Experiment

**Exercise 3.**

Let  $X_1, X_2, X_3, X_4$  equal the cholesterol level of a woman under the age of 50, a man under 50, a woman 50 or older, and a man 50 or older, respectively. Assume that the distribution of  $X_i$  is  $N(\mu_i, \sigma^2)$ ,  $i = 1, 2, 3, 4$ .

(a) Define null and alternative hypotheses and give a critical region for an  $\alpha = 0.05$  significance level.

(b) Construct an ANOVA table and state your conclusion, using the following data:

|         |     |     |     |     |     |     |     |
|---------|-----|-----|-----|-----|-----|-----|-----|
| $X_1$ : | 221 | 213 | 202 | 183 | 185 | 197 | 162 |
| $X_2$ : | 271 | 192 | 189 | 209 | 227 | 236 | 142 |
| $X_3$ : | 262 | 193 | 224 | 201 | 161 | 178 | 265 |
| $X_4$ : | 192 | 253 | 248 | 278 | 232 | 267 | 289 |

(c) Give bounds on the  $p$ -value for this test.

**Exercise 4.**

A pharmaceutical manufacturer wants to investigate the bioactivity of a new drug. A completely randomized single-factor experiment was conducted with three dosage levels, and the following results were obtained:

| Dosage | Observations |
|--------|--------------|
| 20g    | 24 28 37 30  |
| 30g    | 37 41 31 35  |
| 40g    | 42 47 52 38  |

(a) Is there evidence to indicate that dosage level affects bioactivity? Use  $\alpha = 0.05$ .

(b) If it is appropriate to do so, make comparisons between the pairs of means. What conclusions can you draw?

(c) Analyze the residuals from this experiment and comment on model adequacy.